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Bagbazar, Kathmandu

A Case Study Report on

Online Passport Application System:

Department of Passport (DoP), Government of Nepal

E-Governance

(Subject Code: BIT452)

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ABSTRACT

This case study examines the Online Passport Application System developed and operated by the Department of Passports (DoP) under the Government of Nepal. The system represents a significant digital transformation initiative aimed at replacing the traditional, paper-based passport issuance process with a fully integrated, web-based e-governance platform.

The study analyzes the system through the lens of the G2C (Government-to-Citizen) e-governance model, evaluating how the Department of Passports delivers identity documentation services digitally to Nepali citizens, including citizens residing abroad. Key areas of investigation include the functional and non-functional requirements of the system, its architectural design, process workflows, stakeholder roles, and the entity-relationship structure underlying its database.

The findings indicate that the Online Passport Application System has substantially reduced processing time, minimized physical visits to government offices, and increased transparency for applicants. However, challenges persist in areas such as digital literacy, server infrastructure, and equitable access in rural regions. The case study concludes with recommendations for further enhancement aligned with Nepal's broader Digital Nepal Framework.

Keywords: *E-Governance, G2C Model, Passport Application, Department of Passports Nepal, Digital Government, Online Services, Process Flow, ER Diagram*

INTRODUCTION

The Department of Passport (DoP) in Nepal introduced an online passport application system to modernize and streamline the process of applying for, renewing, and managing passports. Traditionally, applicants had to visit passport offices, stand in long queues, submit multiple paper documents, and often face delays in processing. The online system was designed to improve efficiency, transparency, and accessibility for citizens by allowing them to submit applications, upload required documents, and track the status of their passports through digital platforms. The Department of Passport currently operates under the Ministry of Foreign Affairs, Government of Nepal, which is responsible for managing passport services and facilitating international travel for Nepali citizens.

Nepal's Department of Passport (DoP) plays a vital role in issuing passports and ensuring secure identity verification for citizens traveling abroad. As an important governmental body, the department faces several challenges such as handling a large number of applicants, maintaining secure records, preventing identity fraud, and ensuring timely passport issuance. With the growing demand for international travel, the need for an efficient and technology-driven passport management system has become increasingly important.

The Department of Passport was established to provide reliable and secure E-passport services to Nepali citizens. According to the legal provisions of the Passport Act, 2024 (1967) and Passport Regulations, the department has been responsible for issuing, renewing, and managing passports for eligible citizens. Over the years, the department has focused on improving the efficiency, security, and accessibility of passport services. This case study explores the various aspects of the online passport application system, including its implementation, benefits, challenges, and its overall impact on citizens

Background

The Department of Passports (DoP) is a central government agency operating under the Ministry of Foreign Affairs (MoFA), Government of Nepal. It is the authorized body responsible for issuing passports to Nepali citizens within the country and

through diplomatic missions abroad. The main responsibility of the department is to provide secure, reliable, and internationally recognized travel documents for citizens.

Before the introduction of the online passport application system, the passport application process in Nepal was largely manual and time-consuming. Applicants had to visit the passport office multiple times, submit physical documents, and wait in long queues for verification and processing. This traditional system created delays, inefficiencies, and difficulties for citizens seeking passport services.

With the growing demand for passports due to foreign employment, higher education abroad, tourism, and migration opportunities, the need for a more efficient system became essential. To address these issues, the Department of Passports introduced a digitally supported passport management system that allows citizens to apply online, submit required information, and manage their applications more efficiently. This modernization has helped improve transparency, reduce paperwork, and enhance overall service delivery for Nepali citizens.

Objectives

The Online Passport Application System is an important e-governance initiative aimed at modernizing passport services in Nepal by replacing traditional manual procedures with a digital platform. The system focuses on improving service delivery, increasing transparency, and making passport services more accessible to citizens.

Key Objectives of the Online Passport Application System:

1. **Enhancing Efficiency and Reducing Processing Time:** The system aims to speed up the application process and reduce long waiting lines at the Department of Passports and district administration offices.

2. **Improving Security and Reducing Fraud:** Online pre-enrollment is part of a secure, biometric-based system (e-passport) that features an electronic chip to store biometric data (photograph, fingerprints, and retina scans), thereby reducing forgery and fraud.

3. **Standardization with Global Standards:** The system ensures that Nepali travel documents comply with International Civil Aviation Organization (ICAO) standards for security and machine readability.

4. **Convenience for Citizens**: The system allows citizens to fill out forms and schedule appointments online from anywhere, promoting a citizen-friendly service.

5. **Digital Record Keeping**: It allows for the safe and electronic record-keeping of passport information.

6. **Accessibility for Foreign-Based Citizens**: It allows Nepali citizens living abroad to apply for renewals or new passports through online pre-enrollment at Nepali embassies or consulates.

Stakeholders

The successful implementation and operation of the Online Passport Application System involve several stakeholders who interact with the system at different levels.

1. **Citizens (Applicants):** Citizens are the primary users of the system. They use the online portal to register, fill out passport application forms, book appointments, upload required documents, and track the status of their passport applications.

2. **Passport Office Staff:** Passport office staff are responsible for verifying the submitted documents, collecting biometric information such as fingerprints and photographs, and assisting applicants during the passport processing stages.

3. **Passport Officers:** Passport officers review applications, verify applicant details, and approve or reject passport requests according to the rules and regulations set by the government.

4. **System Administrators:** System administrators manage the technical aspects of the system. Their responsibilities include maintaining the database, ensuring system security, updating the online portal, and solving technical problems.

5. **Government Authorities:** The Ministry of Foreign Affairs (MoFA) supervises the Department of Passports (DoP) and ensures that the passport system follows national policies, security standards, and e-governance initiatives.

6. **Nepali Diplomatic Missions Abroad:** Nepali embassies and consulates abroad also act as stakeholders because they assist Nepali citizens living in foreign countries with passport issuance, renewal, and related services through the system.

SYSTEM IMPLEMENTATION

The Online Passport Application System was developed as a web-based platform that can be accessed by citizens through the internet. The system allows users to create an account, fill out the passport application form, upload necessary documents, and book an appointment for biometric data collection and verification. After submitting the application, applicants receive a confirmation with their application details and appointment schedule for visiting the passport office. The system also helps officials manage and verify applicant information efficiently, reducing errors and the chances of fraudulent applications.

E-Governance Model

The Online Passport Application System follows the Government-to-Citizen (G2C) E-Governance Model. In this model, government services are delivered directly to citizens through digital platforms.

Through this system, citizens can interact with the Department of Passports online to apply for new passports, renew existing ones, replace lost passports, and track the status of their applications. The system reduces the need for repeated physical visits to passport offices and simplifies the overall service delivery process.

The G2C model improves transparency, efficiency, and accessibility of government services. By enabling citizens to access passport-related services through the internet, the system strengthens trust in government institutions and supports the development of digital governance in Nepal.

System Features

The Online Passport Application System follows a Government-to-Citizen (G2C) model, allowing applicants to complete most of the passport application process online before visiting the passport office. The system provides several features designed to improve convenience, efficiency, and user experience.

- **User Registration and Login:** Applicants create an account on the passport portal using their personal information and verify their identity through mobile number or email authentication.

- **Online Application Form:** Applicants fill out a detailed digital form where they enter personal details, citizenship information, address, and other necessary data required for passport issuance.
- **Appointment Scheduling:** After submitting the application, the system generates a reference number and allows applicants to book a specific date and time to visit the passport office for biometric data collection such as photograph, fingerprints, and signature verification.
- **Passport Renewal Request:** Existing passport holders can log in to the system and apply for renewal of their passports through the online platform.
- **Lost or Damaged Passport Replacement:** The system allows citizens to apply for replacement if their passport is lost, stolen, or damaged by submitting a request and required supporting documents.
- **Application Status Tracking:** Applicants can track the progress of their passport application online using their reference number or login account.
- **Support and FAQs:** The portal provides a help section and frequently asked questions (FAQs) to guide users and answer common queries related to passport applications and services.

SYSTEM REQUIREMENTS

Functional Requirements

Functional requirements define the essential operations that the Online Passport Application System must perform to support the application, issuance, and management of passports effectively. These requirements ensure that the system meets the needs of both citizens and government officials while maintaining efficiency, accuracy, and security.

Key functional requirements include:

- **User Registration and Login:** Citizens must be able to create an account, log in securely, and manage their profile.
- **Passport Application Submission:** Users can submit new passport applications online, including uploading necessary documents and photographs.
- **Application Tracking:** Citizens can track the status of their applications in real-time.
- **Payment Gateway Integration:** Secure online payment for passport fees.
- **Appointment Scheduling:** Users can book appointments for in-person verification if required.
- **Renewal and Replacement Services:** Support for renewing expired passports or replacing lost/damaged passports.
- **Notification System:** Automatic notifications via email/SMS for application updates, status changes, and reminders.
- **Admin Management:** Government officials can verify applications, approve/reject requests, and generate reports.
- **Verification and Validation:**Validate applicant data against government records and other integrated databases to prevent errors or fraud.
- **Support and Help Services**:Offer a support section and FAQs to assist users in completing the application process.

Non-Functional Requirements

Non-functional requirements define how the system performs rather than what it d oes. These requirements focus on system quality attributes, usability, and operational standards to ensure the Online Passport Application System is reliable, secure, and user-friendly.

- **Security:** The system must protect user data through secure authentication, encrypted communication (e.g., SSL/TLS), and restricted access to sensitive information. Only authorized personnel should have access to applicant records and administrative functions.
- **Availability:** The system should be accessible to citizens at all times via the internet, except during planned maintenance periods. Downtime should be minimized to ensure continuous service delivery.
- **Performance:** The system must handle a large number of concurrent users without delays, especially during peak periods such as passport renewal seasons or deadlines for foreign employment.
- **Scalability:** The platform should be able to expand its capacity and resources as the number of applicants grows, supporting future increases in demand without compromising performance.
- **Usability:** The system interface should be intuitive, user-friendly, and easy to navigate, allowing individuals with limited technical knowledge to apply for passports efficiently.
- **Legal Compliance:** The system must operate in accordance with the **Passport Act, 2024 (1967)**, Passport Regulations, and other relevant government policies related to digital governance, data protection, and international travel documentation standards.

SYSTEM DESIGN

Process Flow Diagram

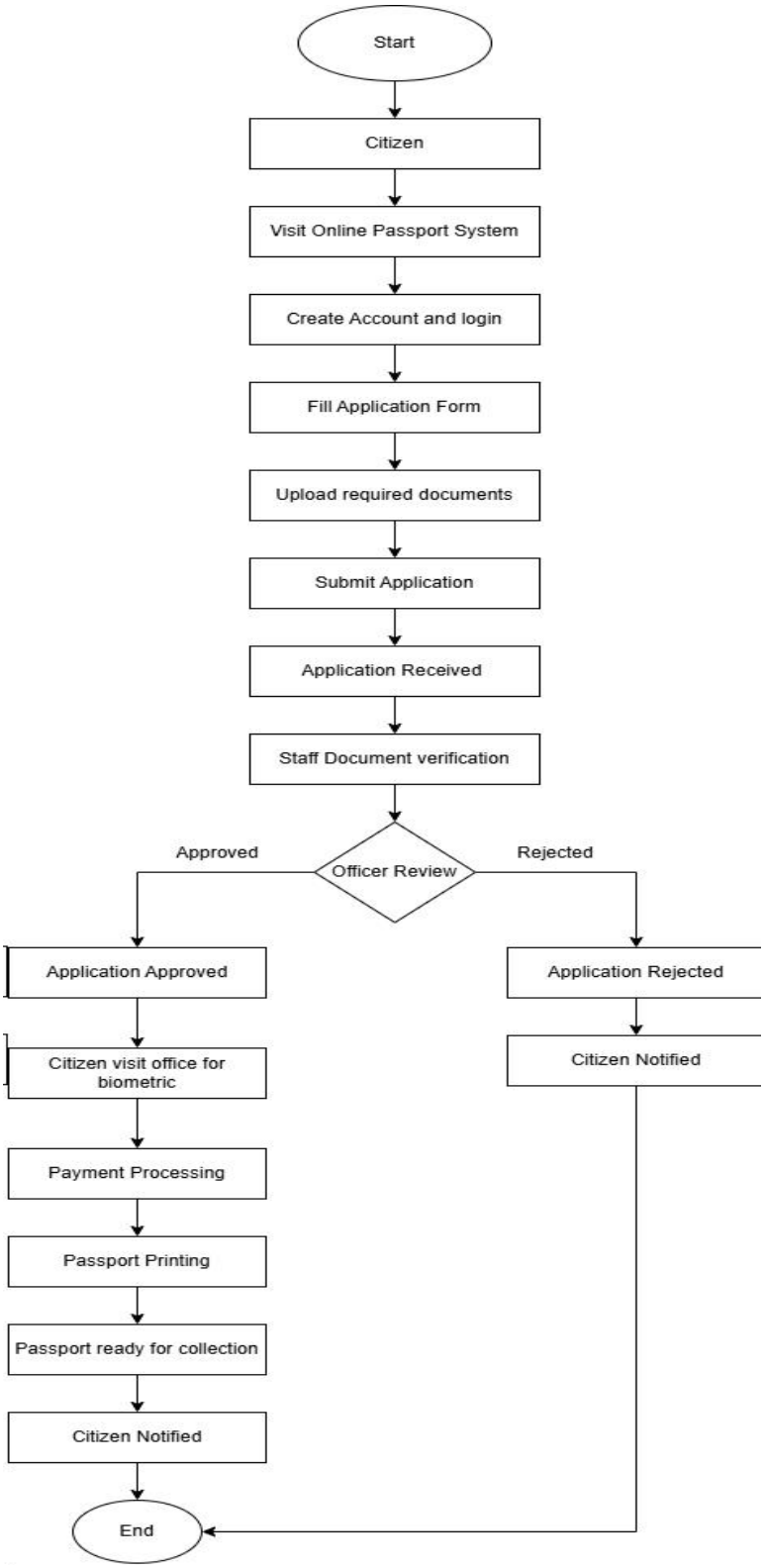

Figure 1: Process Flow Diagram for Online Passport Application

The Process Flow diagram for the Online Passport Application System illustrates the step-by-step workflow that citizens follow to apply for a passport through the Department of Passport's online portal. The process begins when a citizen visits the official online passport system and registers a new account or logs into an existing one. After successful authentication, the applicant fills out the passport application form with the required personal details, uploads all necessary supporting documents, and submits the application. This entire initial phase is completed digitally, making it convenient and accessible for citizens without the need to visit any office at the outset.

Once the application is submitted, the workflow transitions into the government processing phase handled by the Department of Passport. Staff members first verify the uploaded documents, after which a designated officer conducts a thorough review of the application. At this critical decision point, if the application is approved, , the citizen is required to visit the designated office in person for biometric data collection, followed by payment processing, passport printing, and finally the passport is made ready for collection with the citizen notified upon completion, bringing the process to a successful end. However, if the officer determines that the application does not meet the required criteria and rejects it, the citizen is immediately notified of the rejection, and the process concludes at that point without further steps. This structured flowchart ensures that both document integrity and physical biometric verification are completed before a passport is officially issued, maintaining a secure and accountable passport issuance system.

Entity Relationship (ER) Diagram

The **Online Passport Application System** uses a database to store and manage applicant data, application records, officer verification details, document uploads, payment information, and passport generation records. The Entity Relationship (ER) model helps define how different entities in the system interact with each other to ensure a smooth and organized passport issuance process.

Main Entities

- **User (Applicant):** This entity stores the personal information of citizens who apply for a passport through the online portal, including full name, gender,

date of birth, citizenship number, National Identity Document (NID), email, phone number, and password.

- **Application:** This entity represents the passport application submitted by the user, capturing details such as passport type, submission date, and fee paid, serving as the central entity that connects all other components of the system.
- **Officer:** This entity represents the passport department officers responsible for reviewing and processing applications, storing attributes such as officer ID, name, designation, role ID, and the associated department.
- **Department:** This entity stores information about the government department overseeing the passport process, including department name and location, and is linked to the officers working within it.
- **Document:** This entity records all supporting documents uploaded by the applicant, including document type, file path, and uploaded date, linked to the application via a foreign key.
- **Payment:** This entity stores payment information related to the passport application fee, including payment ID, amount, payment method, payment date, and the associated application ID.
- **Status:** This entity tracks the current status of each application, storing a status ID and status name such as pending, approved, or rejected.
- **Passport:** This entity represents the final issued passport, storing details such as passport ID, passport type, issue date, expiry date, and the linked application ID as a foreign key.

Relationships Between Entities

- A **User** can submit multiple **Applications**, forming a one-to-many relationship.
- Each **Application** is associated with one **User**, forming a many-to-one relationship.
- Each **Application** has multiple **Documents** uploaded against it, forming a one-to-many relationship.
- Each **Application** is processed and reviewed by one **Officer**, forming a many-to-one relationship.
- Each **Application** has one associated **Payment** record for the passport fee, forming a one-to-one relationship.

- Each **Application** has one **Status** record reflecting its current processing state, forming a one-to-one relationship.
- Each approved **Application** generates one **Passport**, forming a one-to-one relationship.
- A **Department** has multiple **Officers** working under it, forming a one-to-many relationship.

These relationships ensure proper data organization and allow the system to efficiently track applications, document uploads, payment transactions, officer reviews, and the final passport generation process from start to finish.

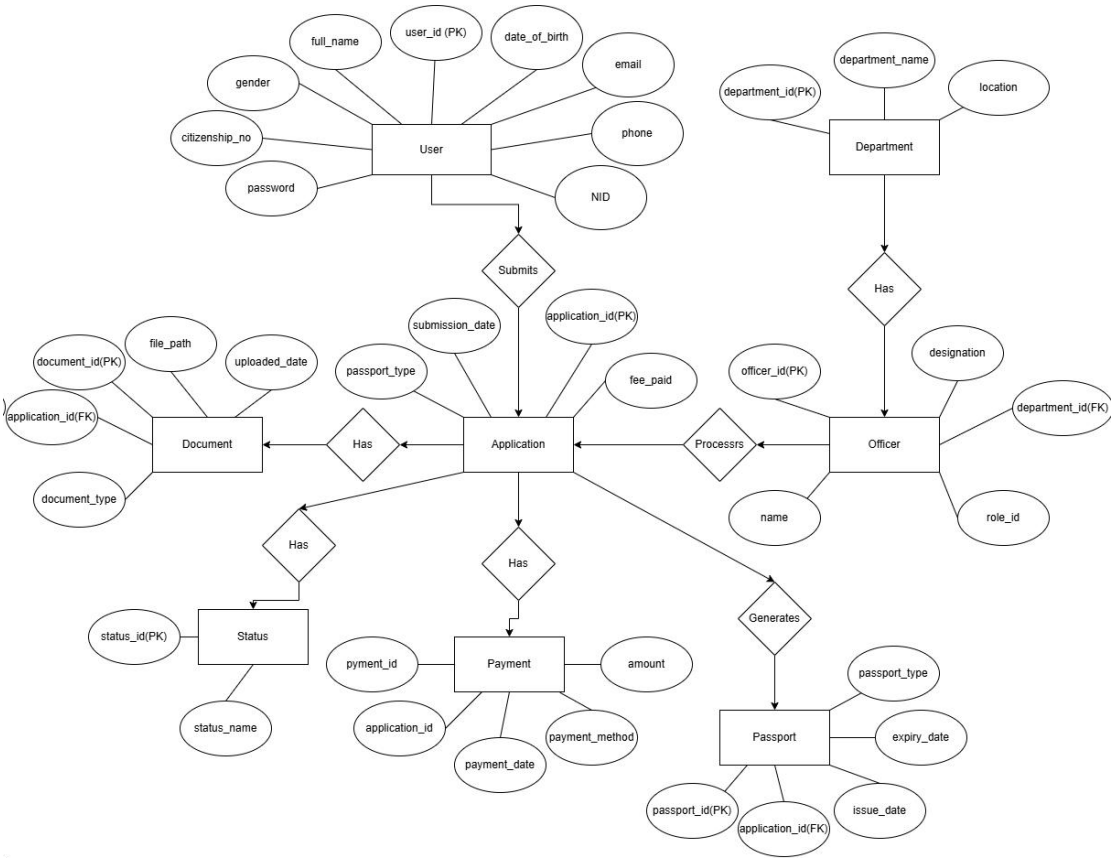

Figure 2: ER Diagram of Online Passport Application

The **Entity Relationship (ER) diagram for the Online Passport Application System** illustrates the database structure and relationships between eight main entities: User, Application, Officer, Department, Document, Payment, Status, and Passport. The **User** entity stores personal information of citizens who register on the online passport portal, including their full name, gender, date of birth, citizenship number,

National Identity Document (NID), email, phone number, and password. Each user can submit multiple Applications over time, creating a one-to-many relationship between User and Application, with the **Application** entity capturing key details such as passport type, submission date, and fee paid, serving as the central entity that connects all other components of the system.

The ER diagram further illustrates how applications are managed, verified, and fulfilled through interconnected entities. Each Application is processed by one **Officer**, while a single officer can handle multiple applications, establishing a many-to-one relationship between Application and Officer. Officers are organized under a **Department**, which stores the department name and location, forming a one-to-many relationship between Department and Officer. Every application must have at least one associated **Document** record containing the uploaded supporting files, and exactly one **Payment** record for the passport fee, capturing the amount, payment method, and payment date. Each application also carries a **Status** record reflecting its current processing state, such as pending, approved, or rejected. Finally, upon successful approval, each application generates exactly one **Passport** record storing the passport type, issue date, and expiry date, creating a one-to-one relationship between Application and Passport. This structured data model ensures proper tracking of applicants, document submissions, officer assignments, payment transactions, and passport generation throughout the entire passport issuance process.

ANALYSIS

Benefits of the Online Passport Application System

The introduction of the Online Passport Application System by the Department of Passport in Nepal has brought about numerous significant benefits for both citizens and government officials:

- **Convenience:** Applicants can complete the entire initial application process from the comfort of their homes by registering an account, filling out the application form, uploading required documents, and scheduling appointments online, saving considerable time and effort without the need for repeated physical visits to the passport office.
- **Transparency:** The digital process reduces opportunities for corruption and ensures a fair, accountable, and transparent application procedure, as all submissions, verifications, and decisions are recorded and tracked within the system.
- **Efficiency:** The online system eliminates the need for citizens to queue for hours simply to submit an initial application. Automation of the application and document verification process speeds up the overall processing timeline, reducing delays in passport issuance significantly.
- **Error Reduction:** Digital data entry and automated validation minimize human errors commonly associated with manual paperwork, ensuring greater accuracy in applicant records and passport details.
- **Environmental Impact:** A significant reduction in paper-based documentation and physical forms contributes to environmental sustainability by lowering the overall paper consumption of the Department of Passport.
- **Security:** The introduction of E-passports embedded with electronic chips storing biometric data such as photographs, fingerprints, and retina scans prevents forgery and identity fraud, while also complying with International Civil Aviation Organization (ICAO) standards for secure and machine-readable travel documents.
- **Accessibility for Foreign-Based Citizens:** Nepali citizens residing abroad can conveniently apply for passport renewal or replacement through online pre-enrollment at Nepali embassies and consulates, removing the need to return to Nepal for routine passport services.

Critical Challenges

Despite its significant goals and benefits, the Online Passport Application System faces several important implementation barriers that must be addressed to ensure equitable and effective service delivery:

- **Processing Backlog:** Significant delays exist in passport printing and distribution, with a large number of applicants experiencing extended waiting periods for their physical passports, often being required to travel on temporary receipts or acknowledgment slips while their passports remain in the printing queue.
- **Infrastructural Gaps:** Poor internet connectivity in rural and remote areas of Nepal, combined with outdated hardware and limited technical infrastructure in some district offices, hinders the nationwide uniform implementation and accessibility of the online passport system.
- **Digital Literacy:** Low digital literacy rates and limited IT skills among a significant portion of the general population make the online portal difficult to navigate independently, leading many applicants to rely on third-party assistance or cyber café operators, which raises concerns about data privacy and security.
- **System Stability:** The online portal has experienced frequent technical disruptions, slowdowns, and system outages, particularly during high-traffic periods such as peak foreign employment seasons or following major policy changes, negatively affecting the user experience and service reliability.
- **Biometric Verification Dependency:** Despite the online application process, citizens are still required to physically visit the passport office for mandatory biometric data collection, which continues to create congestion and long waiting times at passport offices, partially offsetting the convenience offered by the digital system.
- **Cybersecurity Threats:** As the system stores highly sensitive personal and biometric data of citizens, it remains a potential target for cyberattacks and data breaches, requiring continuous investment in advanced security measures and regular system audits to maintain data integrity and public trust.

CONCLUSION AND RECOMMENDATION

The analysis of the Online Passport Application System was conducted to understand the existing passport application process, identify its core limitations, and determine the requirements necessary for developing an efficient digital solution. The traditional passport application process in Nepal required citizens to physically visit the Department of Passport multiple times, submit paper documents, and wait in long queues for verification and processing. This manual approach resulted in significant delays, excessive paperwork, lack of transparency, and considerable difficulties for citizens living in remote areas who had to travel long distances simply to access passport services.

The proposed Online Passport Application System addresses these challenges by providing a secure, web-based portal that allows citizens to register an account, complete the application form, upload required documents, and track their application status entirely online under the Government-to-Citizen (G2C) e-governance model. The system integrates key functionalities including appointment scheduling, biometric data collection, payment processing, and automated notifications, significantly reducing the need for repeated physical visits to the passport office. Government officers are also equipped with a dedicated dashboard to efficiently review, verify, approve, or reject applications, improving overall departmental productivity.

From a requirements perspective, the system must fulfill both functional and non-functional requirements to operate effectively. Functionally, the system must support user registration, online form submission, document uploads, real-time status tracking, secure payment integration, and automated notifications. Non-functionally, the system must ensure high security through encrypted communication and authentication, maintain consistent availability and performance during peak periods, and comply with the Passport Act, 2024 and relevant ICAO international standards. Together, these requirements form the foundation for a reliable, scalable, and citizen-friendly passport management system.

To further enhance the system's impact and address the identified challenges, the following recommendations are proposed:

- **Mobile Application Development:** A dedicated mobile application should be launched for both Android and iOS platforms to provide easier and more convenient access to passport services, particularly for citizens in remote areas of Nepal with limited access to desktop computers or laptops.
- **Multi-Language Support:** The online portal should incorporate Nepali and other local languages to significantly improve accessibility and usability for citizens with limited English proficiency, ensuring that language barriers do not prevent any eligible citizen from applying for a passport independently.
- **Integration with Other Government Agencies:** The passport system should be connected with citizenship databases, the National ID management system, civil registration records, and other relevant government databases to enable real-time data verification, reduce document fraud, and streamline the applicant identity validation process.
- **AI-Powered Appointment Scheduling:** An intelligent, AI-driven appointment scheduling system should be implemented to optimize officer workloads, distribute applicant traffic evenly across passport offices, and significantly reduce waiting times during peak application periods.
- **Digital Payment Expansion:** Multiple payment gateways should be integrated into the system, including popular mobile wallets such as eSewa and Khalti, as well as internet banking options, to provide greater payment convenience and flexibility for applicants from all economic and geographical backgrounds.
- **Suggested Appropriate E-Governance Model:** For the Online Passport Application System, the **G2C2G (Government-to-Citizen-to-Government) Model** is the most appropriate e-governance model. This is because the system involves a complete two-way interaction — citizens initiate the process by submitting applications, uploading documents, and making payments online, while the government receives, verifies, and responds with an approval or rejection decision. The interdepartmental data validation against national citizenship and NID records further strengthens the G2C2G nature of the system. The feedback loop is completed when the government officially notifies the citizen and issues the passport, making it a full G2C2G cycle that best reflects the bidirectional and interdepartmental nature of the entire passport issuance workflow.

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